Who Loses More from Job Displacement? Gender Differences in South Korea

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Abstract

This paper examines gender differences in the labor-market effects of job displacement among married workers in South Korea. Using population-wide administrative data, I compare workers displaced by plant closures with matched non-displaced workers and estimate dynamic effects on employment and earnings. I distinguish raw gender differences from composition-adjusted differences by reweighting women to the male distribution of pre-displacement job characteristics. Job displacement generates persistent losses for both married women and men, but women experience larger losses after accounting for pre-displacement job characteristics. In the composition-adjusted comparison, women's losses are 4.6 percentage points larger than men's in ever work and 11.0 percentage points larger in relative earnings. The presence of young children further amplifies these gender differences. These findings provide new evidence from Korea on the gendered consequences of job displacement among married workers.

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1 Introduction

Job displacement is a major labor market shock with long-lasting consequences for workers. A large literature documents that job displacement leads to substantial and persistent losses in employment and earnings (Davis and von Wachter, 2011; Jacobson et al., 1993; Lachowska et al., 2020; Ruhm, 1991; Schmieder et al., 2023). However, much of this evidence is based on male workers, and relatively little is known about how post-displacement outcomes differ between women and men. A small literature examines gender differences in employment, earnings, and adjustment after job loss (Maxwell and D'Amico, 1986; Crossley et al., 1994; Kunze and Troske, 2015; Meekes and Hassink, 2022). More recent studies extend this line of work by accounting for pre-displacement differences in workers' labor market characteristics and examining the sources of gender gaps in post-displacement recovery (Ivandić and Lassen, 2023; Illing et al., 2024). These studies show that women experience larger relative losses after displacement, with parenthood playing an important role in shaping these gaps. Yet evidence outside Western labor markets remains limited. Evidence from South Korea is informative because gender gaps in labor market outcomes remain large and household responsibilities continue to be unevenly distributed between women and men.

In this paper, I examine how job displacement affects employment and earnings among married workers in South Korea and whether these effects differ by gender. I use job losses induced by plant closures as plausibly exogenous labor market shocks. Following Illing et al. (2024), I distinguish between raw gender gaps and composition-adjusted gender gaps in post-displacement outcomes. The raw gap captures how the average consequences of job displacement differ between married women and men, given their observed positions in the labor market. The composition-adjusted gap asks whether gender differences in post-displacement losses persist when women and men are made comparable in terms of their pre-displacement labor market characteristics, including age, prior earnings, tenure, industry, and firm size. I then examine whether gender differences vary with the presence and age of children. This dimension is particularly important for married workers because job loss is not only an individual labor market shock but also a household-level shock, and children may shape the constraints under which labor-supply decisions are made.

To conduct this analysis, I combine four types of confidential administrative microdata from Korea—the Employer–Employee data, the Business Register, the Population Census, and the Child Register —covering the period 2015 to 2023. I convert annual information in these datasets into a quarterly panel that tracks individuals' employment status, job separations, and firm closures with high temporal precision. These newly available Korean administrative data provide a rare opportunity to link workers' labor market histories to firm

closures and family circumstances in a population-wide setting. This linked structure allows for a more precise analysis of how job displacement affects married women and men, and how these effects vary with the presence and age of children.

The empirical strategy compares married workers displaced by plant closures with non-displaced married workers, separately by gender. To construct comparable control groups, I use coarsened exact matching within each gender. Displaced and non-displaced workers are matched on pre-displacement characteristics such as age, tenure, industry, establishment size, firm revenue, and quarterly earnings. A dynamic difference-in-differences event-study specification is then estimated on the matched sample, tracing workers' employment and earnings from four quarters before displacement to twenty quarters afterward. Gender-specific estimates from this model are used to construct raw gender gaps. Composition-adjusted gaps are obtained by reweighting the female sample so that its distribution of pre-displacement labor market characteristics is comparable to that of the male sample. These characteristics include age, tenure, quarterly earnings, industry, and establishment size. The analysis then examines whether the remaining gender gap after reweighting varies by the presence of children. It further compares post-displacement losses by the age of the youngest child to assess whether gender differences are more pronounced when childcare constraints are likely to be stronger.

The results show that job displacement generates persistent labor market losses for both married women and men. In the unadjusted comparison, women experience larger initial losses than men along several margins, although some of these gender differences narrow over time. Once women are reweighted to have pre-displacement job characteristics comparable to those of men, however, a much larger gender gap emerges. For ever work, the average post-displacement loss is 4.8 percentage points for men and 7.2 percentage points for women in the unadjusted comparison, but rises to 9.4 percentage points for reweighted women. This nearly doubles the female-male gap, from 2.4 to 4.6 percentage points. A similar pattern appears for relative earnings: earnings fall by 8.7% of pre-displacement earnings for men and 12.0% for women in the unadjusted comparison, while the loss for reweighted women rises to 19.7%. The corresponding gender gap therefore increases from 3.3 to 11.0 percentage points. These results indicate that the unadjusted comparison understates women's displacement losses among workers displaced from comparable jobs. The results further suggest that caregiving responsibilities are one channel through which the composition-adjusted gender gap in displacement losses emerges. Among reweighted women, displacement losses are substantially larger when young children are present: the average post-displacement loss in ever work is 14.0 percentage points for women with a young child, compared with 8.0 percentage points for women without a young child. For men, the pattern goes in the opposite

direction, with losses of 4.0 and 5.4 percentage points, respectively. A similar pattern appears for relative earnings: losses are larger for reweighted women with a young child than for those without one, at 26.1% versus 17.6% of pre-displacement earnings, while the corresponding losses for men are 6.6% versus 10.2%. The results by the age of the youngest child further show that this pattern is stronger when the youngest child is younger. Among reweighted women, displacement losses are largest when the youngest child is very young and decline as children grow older, while men's losses vary much less with child age. The comparison with workers without children suggests that the relevant margin is not parenthood itself, but the caregiving responsibilities associated with very young children.

This paper contributes to the existing literature in two ways. First, it contributes to the literature on gender differences in labor market losses following job displacement by providing new evidence from South Korea. The broader job displacement literature has traditionally relied heavily on male workers, leaving relatively limited evidence on whether displacement losses differ by gender. A small number of earlier studies estimate employment and earnings losses separately for women and men (Maxwell and D'Amico, 1986; Crossley et al., 1994), while subsequent work examines specific margins of gendered adjustment after job loss, such as job search, commuting, and working hours (Kunze and Troske, 2015; Meekes and Hassink, 2022). Closely related to this paper, Ivandić and Lassen (2023) and Illing et al. (2024) provide recent evidence on gender gaps in displacement losses in Denmark and Germany, respectively. They differ from earlier studies by accounting for differences in pre-displacement labor market characteristics and by providing a more detailed analysis of the sources of the gender gap. Ivandić and Lassen (2023) use plant closures in Denmark and show that women face a higher risk of unemployment and larger relative earnings losses after displacement. The unemployment gap remains substantial even after accounting for observable differences between women and men, and childcare accounts for an important part of the gender gap in recovery. Illing et al. (2024) use mass layoffs and plant closures in Germany and show that women experience larger earnings losses than men. This gap becomes even larger after reweighting women to have similar pre-displacement labor market characteristics as men and widens sharply with parenthood. Motivated by the approach in Illing et al. (2024), this paper examines gendered displacement losses in South Korea. Rather than fully replicating their analysis, I focus on whether gender gaps in post-displacement losses persist after accounting for pre-displacement labor market characteristics and whether these gaps vary with the presence and age of children. By focusing on South Korea, this paper extends the still limited evidence on gendered displacement losses beyond Western labor markets. This context is also relevant because gender gaps in labor market outcomes

remain large by OECD standards[1], while childcare and household responsibilities remain unevenly distributed within households[2].

Second, it contributes to the literature on job displacement in Korea. Existing studies on Korea have mainly examined average displacement costs, short-run losses, or job-transition patterns. Compared with the extensive literature on displaced workers in the United States and other Western economies, evidence from Korea remains relatively limited. The small existing literature on Korea has mainly examined average displacement costs, short-run wage losses, or subsequent job-transition patterns (Keum and Cho, 2002; Kim and Choi, 2008; Yoon and Park, 2008). Park (2010) extends this literature by documenting that wage losses persist up to four years after job loss. However, relatively little is known about whether displacement losses differ between women and men in Korea. It also remains unclear whether these gaps persist after accounting for pre-displacement labor market characteristics or vary across household contexts. Using newly available linked administrative data, this paper addresses these gaps by examining gender differences in the employment and earnings effects of job displacement among married workers over the twenty quarters following displacement. By linking workers, firms, households, and children in population-wide administrative data, the paper provides a more detailed analysis of post-displacement losses than was possible in earlier Korean studies. The analysis also distinguishes raw gender gaps from composition-adjusted gaps. It further examines heterogeneity by the presence of children and the age of the youngest child, allowing the analysis to assess whether these gaps are larger in households with greater potential care responsibilities.

The paper proceeds as follows. Section 2 describes the data and the sample construction. Section 3 outlines the empirical strategy. Section 4 presents the main results on how labor market outcomes after job displacement differ by gender. Section 5 concludes.

2 Data

2.1 Korean Administrative Data

This study relies on four confidential administrative datasets from Korea: the Employer–Employee Data, the Business Register, the Population Census, and the Child Register, covering the period from 2015 to 2023. The combined data cover the universe of establishments and

[1]Korea has the largest gender wage gap among OECD countries. The OECD gender wage gap is defined as the difference between median earnings of men and women relative to median earnings of men, based generally on unadjusted gross earnings of full-time wage and salary workers.

[2]According to the 2024 Korean Time Use Survey, women spend 179 minutes per day on housework, compared with 62 minutes for men, implying that women do about 2.9 times as much housework as men. Even among dual-earner couples, wives spend 171 minutes per day on housework, compared with 59 minutes for husbands.

workers in Korea. This structure provides a high degree of precision in tracing individuals' labor market histories, observing household-level details such as spouses and children, and identifying firm closures.

The Employer–Employee Data is an annual panel dataset that includes all job information for the entire population each year. These records are drawn from Korea's administrative systems, including the four major social insurances and the National Tax Service. The dataset contains only basic demographic characteristics, such as age and sex, but provides detailed job-level information. This includes hire and separation dates[3], annual earnings, employment type, industry, and indicators of employer size. One caveat of this dataset is that employer identifiers are assigned at the business-group level rather than at the establishment level. As a result, workers employed at different establishments within the same corporate group are recorded under a common employer identifier. To avoid ambiguity in linking workers to establishment closures, the analysis focuses on closures of single-establishment firms, as discussed in Section 2.2.

The Business Register is an annual census of all domestic establishments in Korea, constructed from a combination of administrative records—such as business registration files—and statistical data sources. For each establishment, the register provides opening and closure dates, industry classification, revenue, and other establishment characteristics. This information is used to complement the Employer-Employee Data, which do not indicate whether and when an employer closes. Linking the two datasets makes it possible to identify establishment closure dates and add firm-level characteristics, such as revenue, to the worker records.

The Population Census is an annual, registry-based dataset constructed from administrative sources, including the resident registration system, foreign resident registration records, and the family relations registry. It covers the universe of domestic and foreign residents living in Korea as of November 1 of each year. Using individual identifiers and relationship-to-household-head codes in the Population Census, I identify the spouse of each married worker when both spouses live in the same household in a given year. This information is used to restrict the sample to workers who are married in both the event year and the year before the event. The spouse identifiers from the Population Census are also linked to the Employer-Employee Data, making it possible to observe the employment status and earnings of sample workers' spouses.

As described above, the empirical analysis is based on an annual panel dataset constructed by linking the four administrative datasets introduced in this section. While the full dataset

[3]Hire and separation dates are recorded in year-month format. If a worker remains employed through the end of a calendar year, the separation date for that year is recorded as December of the same year.

covers the period from 2015 to 2023, employer identifiers in the Employer–Employee Data are available only for 2015–2020. As a result, job displacement due to firm closure can be measured only during this period. Within this window, the analysis is restricted to plant closures occurring between 2016 and 2018 to minimize potential confounding from the COVID-19 pandemic. This restriction allows post-displacement outcomes to be observed for several quarters before pandemic-related labor market shocks occurred. For the empirical analysis, the annual data are further converted into a quarterly panel to exploit the timing of key events more precisely. This conversion is possible because key date variables, including hire dates, separation dates, and plant closure dates, are recorded in year-month format and can therefore be assigned to calendar quarters.[4] The resulting quarterly panel spans 2015Q1 to 2023Q4 and provides a longer effective time series for the empirical analysis.

For the analysis, the sample is restricted to workers who are married to the same spouse in both the event year and the year preceding the event. Each displaced worker is matched one-to-one to a non-displaced control worker who does not experience job loss during the same period. For comparability with Illing et al. (2024), the sample is further restricted to individuals aged 24 to 50 in the quarter before the event, corresponding to birth cohorts 1966–1995. The final treatment sample consists of 46,864 displaced workers, including 24,979 women and 21,885 men. The empirical analysis uses quarterly data covering 2015Q1 to 2023Q4, with displacement events defined based on plant closures occurring between 2016Q1 and 2018Q4.

2.2 Definition of Job Displacement

The existing job displacement literature commonly defines displaced workers as those who leave a stable job at their main employer during a mass layoff or plant closure (Davis and von Wachter, 2011; **?**; Illing et al., 2024; Jacobson et al., 1993; Schmieder et al., 2023; Sullivan and von Wachter, 2009). This definition is intended to capture workers who likely lost their jobs involuntarily. However, because the data do not allow mass layoffs to be identified, displacement in this study is defined exclusively as job loss resulting from plant closure. This restriction is unlikely to compromise the validity of the analysis, as plant closures provide a stricter measure of involuntary job loss. Focusing on plant closures helps reduce worker-level selection concerns relative to mass layoffs, where employers may still choose which workers to retain or dismiss. In contrast, because plant closures affect all workers at the closing establishment, job loss is less likely to reflect selection among individual workers within that

[4]Because these date variables are recorded in year-month format, I assign each year-month value to the corresponding calendar quarter. This procedure allows the annual panel to be converted into a quarterly panel and makes it possible to measure employment status and plant closure status at the quarterly level.

establishment. Plant-closure displacement is therefore more plausibly orthogonal to displaced workers' individual characteristics.

One important restriction is that the plant-closure measure is limited to single-establishment business groups, which allows displacement events to be identified without ambiguity. As discussed in Section 2.1, this restriction follows from the structure of the administrative data: establishment-level closure information can be linked reliably to workers' employment records only in such cases.[5] While this restriction improves the accuracy of identifying involuntary job losses due to plant closure, it inevitably narrows the set of closures analyzed relative to settings where establishment-level closures can be directly identified. As a result, the analysis is limited to workers employed at relatively small, single-establishment firms.[6] Plant closures are further restricted to economically meaningful closures using closure-reason codes recorded in the Business Register. Specifically, the definition includes only closures attributed to business downturns, administrative dispositions, or comparable terminal cessation reasons. Cases associated with mergers, takeovers, seasonal shutdowns, or other non-terminal events are excluded to reduce misclassification and ensure that the measure captures actual closures.

I define a worker as displaced in quarter $t = c$ if their main employment[7] at the closing establishment ends on the recorded closure date or within the three months preceding the closure. To sharpen the treatment definition, workers with multiple plant-closure-related job losses during the sample period are assigned to their first observed event. In addition, I restrict the sample to workers with at least one year of tenure at the establishment in quarter $t = c - 1$, who are employed as regular workers and work at establishments with at least five employees. These restrictions are intended to exclude workers with relatively weak attachment to their employers. While prior studies often impose higher tenure thresholds of two to three years and firm-size thresholds of 30 to 50 employees, these requirements are relaxed in this study to maintain sufficient sample size under the restriction to single-establishment firms.

[5]The Employer-Employee Data report employer identifiers only at the business-group level, while the Business Register contains both business-group and establishment identifiers as well as establishment-level closure dates. For single-establishment business groups, the business-group identifier uniquely maps to an establishment, making it possible to determine whether a worker was employed at an establishment that closed. For multi-establishment business groups, however, the Employer-Employee Data do not identify the specific establishment where each worker was employed. As a result, the data do not reveal whether a worker in such a business group was directly affected by a particular establishment closure.

[6]Appendix Table B.1 compares the characteristics of single- and multi-establishment firms using the 2016 Employment Register and Business Register. Single-establishment firms account for 98% of firms and 57% of workers in 2016, suggesting that they represent a substantial segment of the Korean labor market. As expected, they are smaller in employment size and have a lower share of regular workers than multi-establishment firms.

[7]Main employment is defined as the job with the longest ex-post tenure. This definition is intended to capture a relatively stable employment relationship within the quarter.

Finally, the analysis focuses on plant closures occurring between 2016Q1 and 2018Q4, allowing each worker to be tracked from four quarters before to twenty quarters after displacement.

2.3 Constructing Treatment and Control Group

The analysis sample is constructed in two steps. The first step identifies workers who satisfy the baseline sample restrictions. The second step applies coarsened exact matching (CEM) to construct a comparable control group for displaced workers.

The baseline sample is restricted to workers who satisfy the following conditions. Workers must be aged 24 to 50 at $t = c - 1$; be married to the same spouse in both the event quarter, $t = c$, and one year before the event, $t = c - 4$; be employed at a single-establishment firm with at least five employees at $t = c - 4$; and have at least one year of tenure and hold a regular-worker position at $t = c - 1$.[8]

Then, to construct a comparable control group of non-displaced workers, coarsened exact matching (CEM) is applied without replacement, matching each displaced worker to one control worker. Matching is implemented separately for men and women. Workers are matched on age at $t = c-1$, establishment size at $t = c-4$, industry at $t = c-4$, tenure at $t = c-1$, and quantiles of establishment revenue and quarterly total earnings at $t = c - 4$ (see Appendix A. for details).

Table 1 reports descriptive statistics for displaced and matched non-displaced workers separately by gender. Panel A presents individual characteristics, Panel B presents child status and spouse characteristics, and Panel C presents establishment characteristics. Most variables are measured prior to displacement, while child status is measured at the time of displacement. As shown in Table 1, the analysis sample consists of married workers in their early forties. Men have longer pre-displacement tenure and substantially higher quarterly earnings than women. The table also shows clear gender differences in household characteristics. Male workers are more likely to have young children: 41% have a child under age 8, compared with 21% of female workers. This difference is consistent with lower labor-force attachment among women with young children. Spouse characteristics also differ sharply by gender: female workers have spouses with higher pre-displacement earnings and employment rates. Turning to establishment characteristics, the sample is concentrated among workers employed at smaller establishments, reflecting the focus on closures of single-establishment firms.

[8]The baseline is defined as the quarter immediately preceding the event date, $t = c - 1$. However, because firm-level characteristics, such as revenue, employment size, and industry, are available only annually in the administrative data, the corresponding firm-level baseline information is measured at $t = c - 4$, one year before the event.

Following Austin (2009), covariate balance is assessed using standardized mean differences (SMDs), which provide a scale-invariant measure of between-group differences and are commonly used across different matching methods. The SMD is defined as the difference in means between the two groups divided by the pooled standard deviation, with absolute values below 0.1 generally interpreted as indicating acceptable covariate balance. After matching, the SMDs are close to zero across all covariates for both the female and male samples, indicating that the matched control group is sufficiently similar to the displaced group in observed pre-displacement characteristics. The improvement in covariate balance is also illustrated in Appendix B.1, which plots pre- and post-matching SMDs for each covariate. In the raw (unmatched) sample, several covariates are imbalanced, most notably tenure, quarterly earnings, and revenue. The matching procedure greatly improves balance, reducing all SMDs to levels well within the ± 0.1 threshold. Overall, the matched sample provides a balanced comparison group for both men and women, thereby improving the comparability underlying the subsequent difference-in-differences estimates.

3 Empirical Strategy

3.1 Event Study

The analysis estimates the effects of job displacement on labor market outcomes using an event-study design. This approach traces the evolution of outcomes before and after displacement, thereby capturing the dynamic effects of the shock. Since displaced workers are compared with matched non-displaced workers assigned to the same event date, already-treated cohorts cannot implicitly serve as comparison groups for cohorts displaced at different dates. This design helps avoid the negative- or non-convex-weighting problems that can arise in conventional two-way fixed-effects estimators. The specification is estimated separately for men and women:

$$Y_{itc} = \sum_{l=-4}^{20} \gamma_l \times I[t = c + l] + \sum_{l=-4}^{20} \delta_l \times I[t = c + l] \times D_i + \pi_t + \alpha_i + \epsilon_{itc} \tag{1}$$

Here, Y_{itc} denotes the outcome of interest for individual i observed in calendar quarter t, with event date c. For displaced workers, c corresponds to the quarter of displacement, while for matched non-displaced workers, it is the assigned event date of the matched displaced worker. Event time l is normalized to zero in the quarter of displacement. D_i is an indicator equal to one if individual i is displaced. $I[t = c + l]$ denotes an event-time indicator equal to one if calendar quarter t corresponds to event time l relative to event date

c. The period immediately preceding displacement, $l = -1$, is omitted as the reference period. The coefficients of interest are δ_l, which measure how outcomes for displaced workers evolve relative to those of comparable non-displaced workers at each event time l. The specification includes calendar-quarter fixed effects, π_t, and individual fixed effects, α_i. ϵ_{itc} is the error term. Standard errors are clustered at the company level.

The outcome set includes four labor market outcomes: (i) *Ever work*, an indicator equal to one if the individual holds any job in quarter t, regardless of employment type; (ii) *Regular employment*, an indicator equal to one if the individual holds a job with a contract duration exceeding one month in quarter t; (iii) *quarterly earnings*, defined as total earnings received in quarter t, measured in units of 10,000 KRW and excluding earnings from special-type and freelance employment; and (iv) *relative earnings*, defined as earnings in quarter t divided by the average earnings over the four quarters before displacement.

3.2 Aggregate Difference-in-Differences

To summarize treatment effects over broader time horizons, the analysis also estimates a difference-in-differences specification that aggregates event-time indicators into annual periods relative to the displacement event. This specification is estimated separately by gender:

$$\begin{aligned} Y_{itc} = \beta_0 \left(\mathbf{1}\{\text{During}_t\} \times D_i\right) + \sum_{s=1}^{4} \beta_s \left(\mathbf{1}\{\text{After}_{s,t}\} \times D_i\right) \\ + \gamma_0 \mathbf{1}\{\text{During}_t\} + \sum_{s=1}^{4} \gamma_s \mathbf{1}\{\text{After}_{s,t}\} + \pi_t + \alpha_i + \epsilon_{itc} \end{aligned} \tag{2}$$

where $\mathbf{1}\{\text{During}_t\}$ indicates the year of displacement, and $\mathbf{1}\{\text{After}_{s,t}\}$ denotes an indicator for the s-th year after displacement. The omitted category is the year prior to displacement. The coefficients of interest are β_0 and β_s, which capture the average treatment effects in the year of displacement and in subsequent post-displacement years, respectively. π_t and α_i denote calendar-quarter and individual fixed effects, respectively. ϵ_{itc} is the error term. Standard errors are clustered at the company level.

3.3 Adjusting for Gender Differences: Reweighting Procedure

The empirical analysis assesses gender differences in labor market losses following job displacement. A key challenge in interpreting these differences is that men and women may enter displacement with different pre-displacement characteristics that are themselves related to subsequent labor market losses. As discussed in Section 2.3, Table 1 shows that, even under identical baseline sample restrictions, men and women differ systematically in tenure,

earnings, industry composition, and establishment attributes. Therefore, differences in the estimated displacement effects by gender may reflect not only gender-specific responses to job loss, but also differences in the distribution of characteristics prior to displacement.

To address this concern, I implement a reweighting procedure based on **?**, following recent applications in the job-displacement literature by Ivandić and Lassen (2023) and Illing et al. (2024). The procedure reweights women so that their distribution of pre-displacement characteristics more closely resembles that of men. As a result, women whose pre-displacement characteristics are more similar to those of men receive greater weight in the reweighted sample.

I formalize the comparison using a potential–outcomes framework. Let Y_1 denote the outcome an individual would experience if displaced and Y_0 the outcome that would prevail in the absence of displacement. The individual-level effect of displacement is then defined as $\Delta \equiv Y_1 - Y_0$. With gender indicated by $G \in \{f, m\}$, the raw gender gap in displacement effects is

$$\text{GenderGap}_{\text{unadj}} = E[\Delta \mid G = f] - E[\Delta \mid G = m]$$

Pre-displacement characteristics—such as earnings, industry, and establishment size—are key determinants of the magnitude of displacement effects. Let X denote the vector of these characteristics. Because Y_1 and Y_0 may depend on X, gender differences in the distribution of X can give rise to differences in average displacement effects even in the absence of genuine gender-specific heterogeneity.

To separate these compositional differences from true heterogeneity, I construct the counterfactual expectation of women's displacement effect evaluated at the male covariate distribution:

$$E[E[\Delta \mid G = f, X] \mid G = m] = \int_{\mathcal{X}} E[\Delta \mid G = f, X = x] \, dF_m(x). \tag{3}$$

where $F_m(x)$ denotes the distribution of covariates for men. In practice, $F_m(x)$ is approximated by an estimated reweighting function $\phi(x)$ applied to the empirical distribution of displaced women. The composition-adjusted gender gap is therefore

$$\text{GenderGap}_{\text{adj}} \equiv \int_{\mathcal{X}} E[\Delta \mid G = f, X = x] \, dF_f(x) \, \phi(x) - E[\Delta \mid G = m]. \tag{4}$$

This adjusted measure isolates the portion of the gender difference in displacement effects that persists after accounting for differences in pre-displacement characteristics. The remaining gap reflects gender-specific responses to job displacement rather than differences in covariate distributions.

Among displaced workers, I estimate a probit model in which the dependent variable is an indicator for being male. I use the predicted probabilities to construct the reweighting function $\phi(x)$, which gives greater weight to women whose covariate profiles more closely resemble those of men. The covariates are the same individual and establishment characteristics used in the coarsened exact matching: age at $c-1$, total quarterly earnings at $c-4$, industry at $c-4$, establishment size at $c-4$, and tenure at $c-1$.

From this regression, I obtain the predicted propensity score $\hat{p}$ and construct the reweighting function as $\phi(x) = \hat{p}/(1-\hat{p})$. Table 2 reports descriptive statistics for displaced women, reweighted displaced women, and displaced men at the reference quarter. Before reweighting, displaced women differ substantially from displaced men in age, earnings, and establishment characteristics (Columns 1 and 3). After reweighting, the female sample more closely resembles the male sample, with higher pre-displacement earnings and greater representation in larger and higher-revenue establishments. This adjustment also increases the weight placed on women in industries where they are underrepresented relative to men, as shown in Appendix Table B.3. I apply these weights in both the event-study and aggregate difference-in-differences analyses described in Sections 3.1 and 3.2.

4 Results

4.1 Raw Gender Differences in Displacement Losses

I begin with raw mean trends to illustrate the descriptive patterns underlying the event-study estimates. Figure 1 plots employment and earnings trajectories for displaced and matched non-displaced workers, separately by gender. Panel (a) presents raw mean trends in ever work. In the pre-displacement period, the ever-work rate is mechanically equal to one for all groups.[9] After displacement, employment declines sharply among displaced workers, with a larger drop for women than for men. Although employment partially recovers for both groups, displaced workers do not return to the employment levels of their matched non-displaced counterparts even 20 quarters after displacement. Panel (b) presents raw mean trends in quarterly earnings. A notable feature is the large gender gap in earnings levels even before displacement: men's quarterly earnings are substantially higher than women's for both displaced and non-displaced workers. Within each gender, displaced and non-displaced workers show broadly similar earnings trends before the displacement event. After displacement, earnings also fall sharply and remain persistently below the control-group

[9]This reflects the baseline sample restrictions that workers must have at least one year of tenure and hold a regular-worker position at $c-1$.

level. Overall, the raw mean trends show a broadly similar post-displacement pattern for men and women: sharp declines followed by partial recovery, although the extent of the decline and recovery appears to differ by gender. I next turn to event-study specifications to estimate the extent of these losses more formally, separately by gender.

Figure 2 presents the event-study estimates for four labor market outcomes: ever work, regular employment, quarterly earnings, and relative earnings. The figure reports estimates separately for men, women, and reweighted women. In this subsection, I focus on the unadjusted estimates for men and women. The estimates confirm the patterns observed in the raw mean trends. Displacement leads to sharp declines in all outcomes immediately after job loss, followed by gradual but incomplete recovery.

In Panel (a), the ever-work rate falls immediately after displacement for both men and women, with a larger initial decline for women. In the year of displacement, ever work decreases by 9.6 percentage points for men and 15.3 percentage points for women (Table 3, Column 1). The gender gap in employment losses narrows over time, and the male and female recovery paths become more similar after about 17 quarters. Over the post-displacement period, women experience a statistically significant additional 2.4 percentage-point decline in ever work relative to men (Table 4, Column 1).

Panel (b) shows a similar pattern for regular employment, but the losses are larger and more persistent than those for ever work. In the year of displacement, regular employment decreases by 17.6 percentage points for men and 20.8 percentage points for women (Table 3, Column 2). The initial decline is larger for women, but the gender gap narrows over time as female regular employment recovers somewhat faster than male regular employment. Consistent with this convergence, the interaction estimates indicate that women experience a 1.3 percentage-point smaller decline in regular employment relative to men over the post-displacement period (Table 4, Column 2). This suggests that, unlike ever work, the larger initial loss for women in regular employment does not persist to the same extent over the full post-displacement period.

Panel (c) turns to quarterly earnings. In levels, earnings losses are broadly similar for men and women, and the recovery paths also move closely together over time. However, because men had substantially higher pre-displacement earnings, as shown in Figure 1, comparisons in levels do not fully capture the relative size of earnings losses across genders. Panel (d) therefore examines relative earnings, defined as earnings in each quarter relative to average pre-displacement earnings. On this measure, women experience substantially larger earnings losses than men immediately after displacement, and the gender gap persists throughout the post-displacement period. In the year of displacement, earnings fall by 23.9% of pre-displacement earnings for women, compared with 16.3% for men (Table 3, Column 4). This

gender gap is also evident over the full post-displacement period: the decline in relative earnings is 3.3 percentage points larger for women than for men (Table 4, Column 4).

The results so far show that job displacement generates persistent labor market losses for both men and women, with larger losses for women along some margins, particularly ever work and relative earnings. However, these raw gender differences do not distinguish gender-specific displacement effects from compositional differences in pre-displacement characteristics, as discussed in Section 3.3. The next subsection therefore presents composition-adjusted estimates that reweight female workers so that they are compared to men with comparable pre-displacement job characteristics.

4.2 Composition-Adjusted Gender Differences in Displacement Losses

The estimates in the previous subsection suggest that women experience larger displacement losses than men immediately after job loss, but that these gender differences diminish over time. However, once women are compared to men with comparable jobs using the reweighting procedure described in Section 3.3, a strikingly larger gender gap emerges.

I now turn to the reweighted female estimates in Figure 2. Panel (a) shows the results for ever work. After reweighting, the employment loss for women becomes much larger than in the unadjusted estimates. The reweighted female estimate drops sharply immediately after displacement and remains below both the male and unadjusted female estimates throughout most of the post-displacement period. As shown in the unadjusted estimates, ever work falls by 9.6 percentage points for men and 15.3 percentage points for women in the year of displacement. After reweighting, however, the female estimate implies a larger decline of 17.6 percentage points (Table 3, Column 1). In the average post-displacement estimates, the female-male gap in ever-work losses rises from 2.4 percentage points in the baseline comparison to 4.6 percentage points after reweighting (Table 4, Column 1). Thus, composition adjustment nearly doubles the estimated gender gap in employment losses, suggesting that the unadjusted comparison masks a much larger gap among workers displaced from comparable jobs.

Panel (b) shows that composition adjustment changes the interpretation of regular-employment losses. While the unadjusted estimates suggest that women recover more quickly than men after the initial decline, the reweighted estimates show that women experience larger and more persistent regular-employment losses when compared to men displaced from comparable jobs. This contrast is especially clear in the longer run. Four years after displacement, the baseline comparison suggests smaller regular-employment losses for women

than for men, at 7.5 versus 10.7 percentage points, but after reweighting the female loss rises to 12.8 percentage points, exceeding the male loss (Table 3, Column 2).

A similar pattern appears for quarterly earnings in Panel (c). In the unadjusted estimates, earnings losses are broadly similar for men and women. After reweighting, however, female losses become substantially larger and more persistent than male losses. In the year of displacement, the female earnings loss rises from 1.246 million KRW in the baseline comparison to 2.247 million KRW after reweighting, exceeding the male loss of 1.597 million KRW. The difference persists four years after displacement: the reweighted female loss remains larger than the male loss, at 1.198 million KRW versus 661,000 KRW, whereas the unadjusted female loss is only 372,000 KRW (Table 3, Column 3). This pattern carries over to relative earnings. Although the unadjusted and reweighted female estimates are similar in the first quarter after displacement, they diverge thereafter. The reweighted female losses remain substantially larger throughout the post-displacement period. Averaged over the 20 quarters after displacement, earnings fall relative to pre-displacement earnings by 8.7% for men and 12.0% for women in the baseline comparison. The female loss rises to 19.7% after reweighting, raising the female-male gap from 3.3 to 11.0 percentage points.

Overall, the composition-adjusted estimates show that the baseline comparisons understate women's displacement losses. Once women are compared with men displaced from comparable jobs, women experience larger and more persistent losses in employment and earnings. Appendix Figure B.2 and Figure B.3 show that this pattern is driven primarily by differences in pre-displacement industry and earnings: excluding either variable from the reweighting procedure moves the estimated female losses closer to the unadjusted estimates. This suggests that differences in the types of jobs held by men and women before displacement play an important role in shaping the observed gender gap.

4.3 Heterogeneity by the Presence of Young Children

The composition-adjusted estimates show that, among married workers, women experience larger and more persistent displacement losses than men with comparable pre-displacement job characteristics. One possible reason for this remaining gender gap is that women's post-displacement adjustment may be more constrained by family responsibilities. These constraints are likely to be especially relevant when children are young, since young children require more intensive care and may limit workers' ability to search for jobs, accept jobs with longer hours, or adjust commuting patterns. I therefore examine heterogeneity in the gender gap in displacement losses by the presence of young children.

Figure 3 presents event-study estimates by the presence of young children, focusing on ever work and relative earnings. I define workers as having a young child if their youngest child is below age 8 at the time of displacement. The figure compares men and reweighted women separately by young-child status. The results differ sharply by gender. Panel (a) shows that, among reweighted women, the decline in ever work is much larger for those with a young child than for those without a young child, and their recovery is slower throughout the post-displacement period. For men, however, the difference by young-child status is much smaller and goes in the opposite direction: men with a young child experience slightly smaller losses and recover somewhat faster than men without a young child. This pattern suggests that young children amplify women's employment losses after displacement, while a similar penalty is not observed for men. The average post-displacement estimates in Table 5, Column 1 confirm this pattern. For reweighted women, the ever-work loss is 14.0 percentage points among those with a young child, compared with 8.0 percentage points among those without a young child. For men, the pattern is reversed: the corresponding losses are 4.0 and 5.4 percentage points, respectively. As a result, the gender gap in ever-work losses is much larger among workers with a young child, at 10.0 percentage points, compared with 2.6 percentage points among those without a young child.

Panel (b) shows a similar pattern for relative earnings. For reweighted women, the presence of a young child is associated with a more persistent earnings deficit relative to pre-displacement earnings. For men, the pattern again moves in the opposite direction: men with a young child experience smaller relative earnings losses. Averaged over the post-displacement period, relative earnings fall by 26.1% of pre-displacement earnings for reweighted women with a young child, compared with 17.6% for reweighted women without a young child. For men, by contrast, the corresponding losses are 6.6% and 10.2%, respectively. This implies that the relative-earnings gap between women and men is almost three times as large among workers with a young child as among those without one, at 19.5 versus 7.4 percentage points (Table 5, Column 4). A similar pattern appears for regular employment and quarterly earnings, which are reported in Appendix Figure B.4.

Taken together, these results show that the composition-adjusted gender gap in displacement losses is concentrated among workers with young children. This pattern is consistent with the broader child-penalty literature, which documents that children are associated with persistent declines in women's labor market outcomes but much smaller changes for men (Kleven et al., 2019). In this setting, the results suggest that family responsibilities may be one of the channels that help explain why gender differences in post-displacement adjustment persist even after accounting for pre-displacement job characteristics.

4.4 Displacement Losses by Youngest Child's Age

The preceding results suggest that young children play an important role in shaping the gender gap in displacement losses. To examine this pattern more closely, I estimate displacement losses by the age of the youngest child at the time of displacement. For each youngest-child age, I plot the average post-displacement difference-in-differences estimate separately for men and reweighted women. This analysis asks how displacement losses vary with the age of the youngest child and whether this age profile differs between women and men.

Figure 4 presents the results. I again focus on ever work and relative earnings in the main text, with the corresponding results for regular employment and quarterly earnings reported in Appendix Figure B.5. Panel (a) shows the estimates for ever work. Among reweighted women, displacement losses are largest when the youngest child is very young and become smaller as the child's age increases. The decline is especially pronounced for women whose youngest child is age 0 or 1 at the time of displacement. When the youngest child is age 0, ever work falls by about 19 percentage points for reweighted women, compared with only about 3 percentage points for men. The female losses are concentrated among workers whose youngest child is below age 8; after age 8, the estimated losses for women and men become much more similar. For men, however, the estimates are relatively flat across child ages, indicating that men's employment losses vary little with the age of the youngest child. As a result, the gender gap in ever-work losses is largest among parents with very young children and narrows as children grow older. One notable feature is that reweighted women without children experience larger ever-work losses than women with older children, but smaller losses than women with very young children. This suggests that the gender gap in ever-work losses is driven less by the presence of children itself than by the presence of very young children. For men, consistent with the preceding heterogeneity analysis, workers without children experience slightly larger losses than workers with children, and losses are relatively flat across the age of the youngest child.

Panel (b) turns to relative earnings and shows a similar pattern. Among reweighted women, relative earnings losses are largest when the youngest child is very young. When the youngest child is age 0, relative earnings fall by about 28% of pre-displacement earnings for reweighted women, compared with only about 3.5% for men. The gender gap then narrows as the youngest child becomes older. The estimates for workers without children show a benchmark pattern similar to that in Panel (a): losses for women without children are smaller than those for women with very young children, but remain larger than those for many women with older children. The magnitude of these relative earnings losses is smaller than that reported

by Illing et al. (2024), although the estimates are not directly comparable.[10] For example, when the youngest child is age 0, Illing et al. (2024) report earnings losses of about 60% for women and about 20% for men, whereas the corresponding losses in this paper are about 28% and 3.5%, respectively. One possible reason for the smaller magnitude is the restriction to workers displaced from single-establishment firms. Since displacement losses partly reflect the loss of establishment wage premiums (Schmieder et al., 2023), excluding workers from multi-establishment firms with larger wage premiums or stronger internal career ladders may leave workers with smaller establishment-specific rents to lose before displacement. This could lead to smaller earnings losses relative to pre-displacement earnings than in studies that include workers from both single- and multi-establishment firms.[11]

Overall, the age-profile results reinforce the conclusion that the gender gap in displacement losses is closely linked to the age of the youngest child. For both ever work and relative earnings, losses among reweighted women are largest when the youngest child is very young and decline as children grow older, while the corresponding estimates for men are much flatter. The comparison with workers without children further suggests that the relevant margin is not simply the presence of children, but the presence of young children. These findings sharpen the interpretation of the preceding heterogeneity results: caregiving responsibilities appear to matter most when children are very young, rather than through parenthood itself.

5 Conclusion

This paper examines gender differences in the labor market effects of job displacement among married workers in South Korea. Using population-wide administrative data, I compare married workers displaced by plant closures with matched non-displaced workers and estimate dynamic effects on employment and earnings. By focusing on South Korea, this paper extends the still limited evidence on gendered displacement losses beyond Western labor markets. It also contributes to the Korean context by providing new evidence on gender differences in post-displacement labor market outcomes based on linked administrative microdata.

The findings show that job displacement produces persistent labor market losses for both married women and men, but that the gender gap depends crucially on how women

[10] Illing et al. (2024) measure earnings relative to annual earnings two years before displacement, whereas this paper measures quarterly earnings relative to average pre-displacement earnings. The sample definitions also differ: their couple sample consists of partners who are both in the labor force and covered by social security, while this paper can identify spouses regardless of whether they are employed. Moreover, this paper restricts displacement events to single-establishment firms.

[11] This explanation may be particularly relevant for the male estimates, since men's displacement losses may be more closely tied to establishment wage premiums and internal career ladders. For women, caregiving-related labor-supply adjustments may still generate larger losses when children are very young.

and men are compared. In the unadjusted comparison, women experience larger initial losses along several margins, although some differences narrow over time. After adjusting for pre-displacement job characteristics, however, women experience substantially larger and more persistent losses than men displaced from comparable jobs. This suggests that raw comparisons mask the extent to which women face larger post-displacement losses among workers with similar pre-displacement labor market characteristics.

The analysis further shows that the presence and age of children provide one lens through which the remaining gender gap can be understood. Women's displacement losses are substantially larger when young children are present, whereas for men the pattern is reversed and weaker in magnitude. Losses among women are largest when the youngest child is very young and decline as children grow older, while men's losses vary much less with child age. Notably, the comparison with workers without children suggests that the relevant margin is not parenthood itself, but the caregiving responsibilities associated with very young children.

Future research could examine additional mechanisms underlying gender differences in displacement losses among married workers. This paper highlights children and caregiving responsibilities as one important margin through which gender differences in displacement losses may arise. However, gender differences among married workers are likely shaped by a broader set of mechanisms. Household-level factors, such as spouses' employment, earnings, and labor-supply responses, may influence how workers adjust after displacement. Demand-side factors may also matter, including gender differences in how employers evaluate displaced workers and occupation- or industry-specific reemployment opportunities. Future work that jointly considers these various channels could provide a more complete understanding of why displacement losses differ between married women and men.

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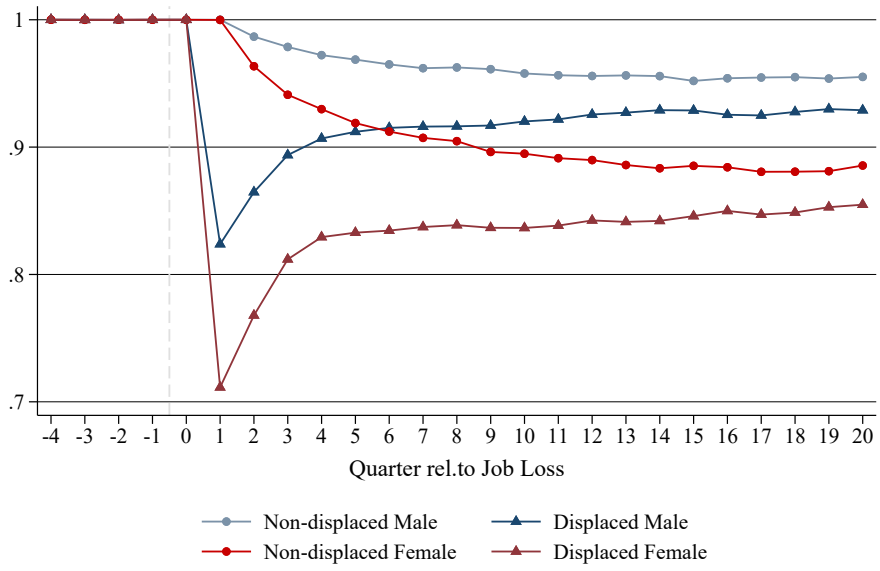
Schmieder, J. F., T. von Wachter, and J. Heining (2023). The costs of job displacement over the business cycle and its sources: Evidence from germany. *American Economic Review 113*(5), 1208–1254.

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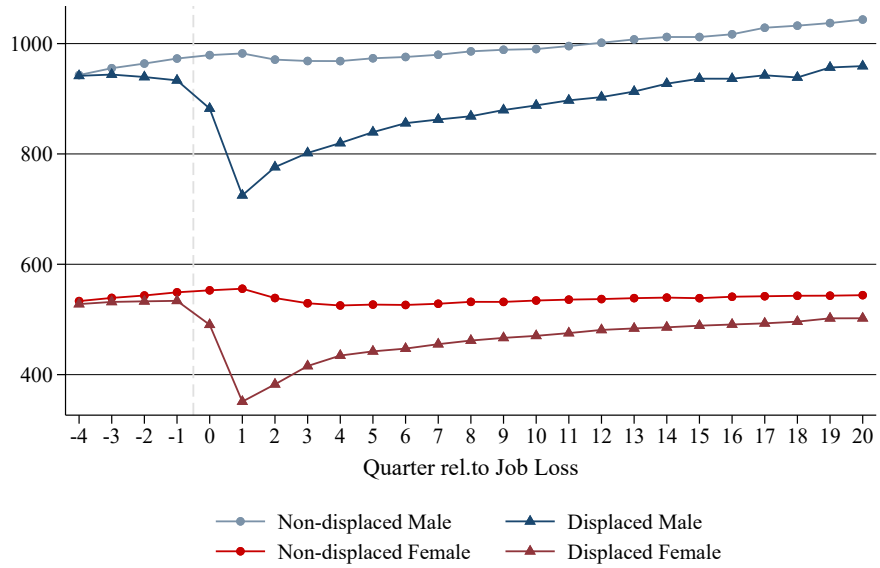
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Figures

Figure 1: Mean Trends in Employment and Earnings Around Job Displacement



(a) Ever work

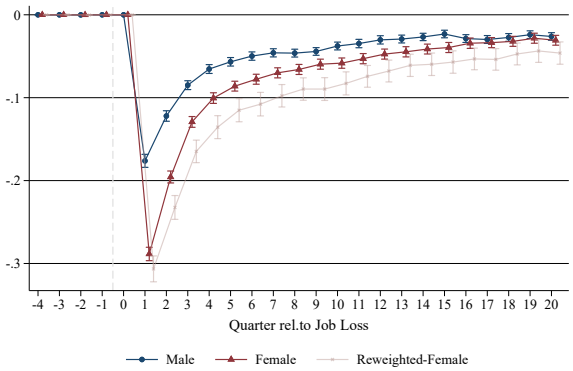


(b) Quarterly earnings

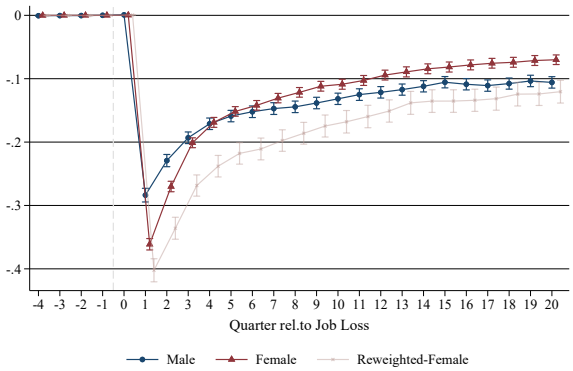
Notes: This figure plots mean trends in ever work and quarterly earnings around job displacement for displaced and non-displaced workers, separately by gender. Triangles indicate displaced workers, and circles indicate non-displaced workers. Red-toned lines correspond to women, while blue-toned lines correspond to men. Quarterly earnings are expressed in units of 10,000 KRW.

Figure 2: Effects of Job Displacement on Labor Supply

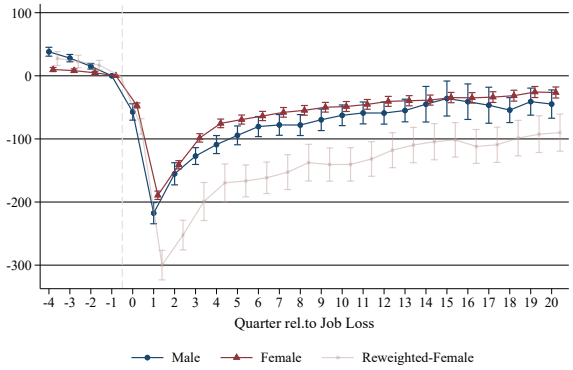
(a) Ever Work



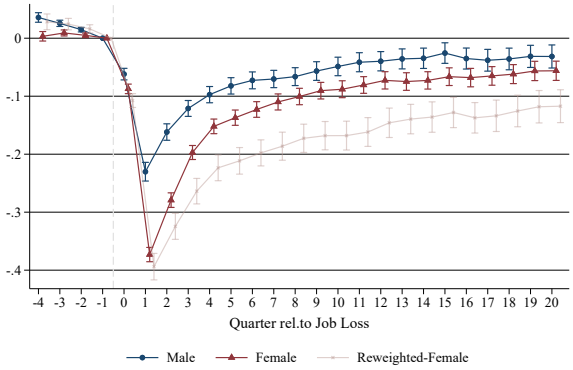
(b) Regular employment



(c) Quarterly Earnings

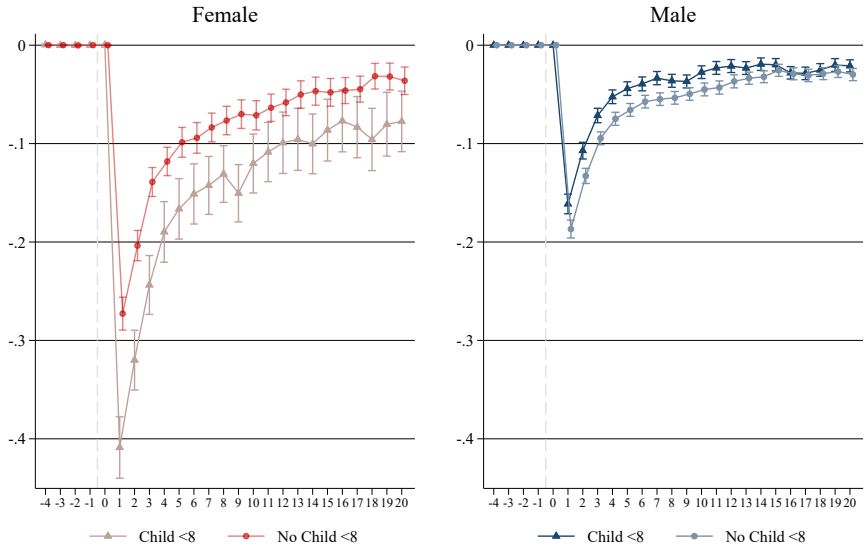


(d) Relative earnings

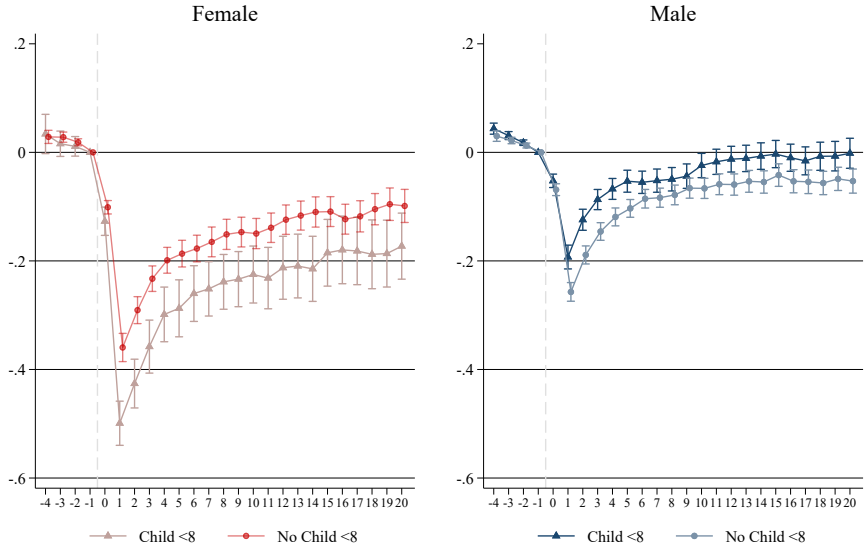


Notes: This figure presents dynamic difference-in-differences estimates. Panel (a) reports effects on employment(including all employment status), Panel (b) on regular employment, Panel (c) on quarterly earnings, and Panel (d) on relative earnings. Quarterly earnings in Panel (c) are expressed in units of 10,000 KRW. The specification includes calendar-time fixed effects and individual fixed effects. Each dot represents the estimated interaction coefficient between treatment status and event time, and capped lines indicate 95% confidence intervals. Standard errors are clustered at the company level.

Figure 3: Effects of Job Displacement on Labor Market Outcomes by the Presence of Young Children



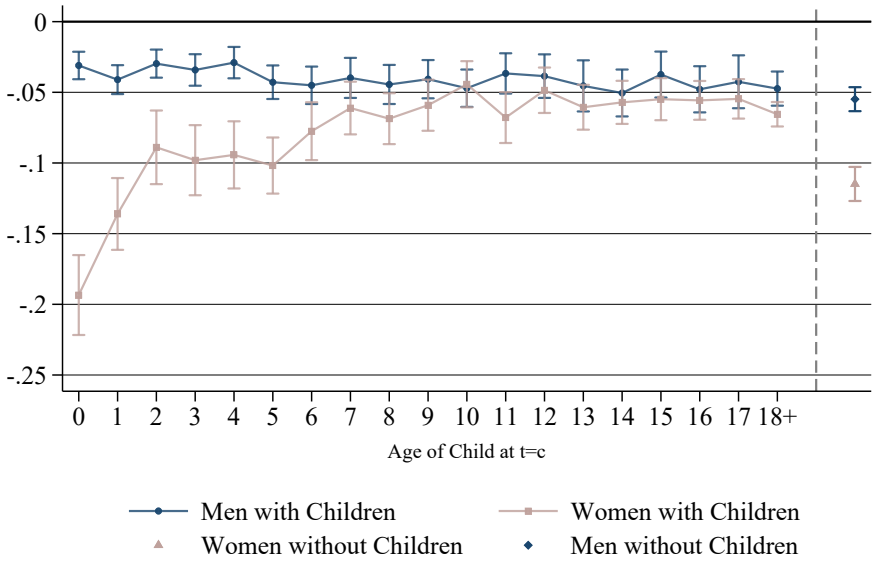
(a) Ever work



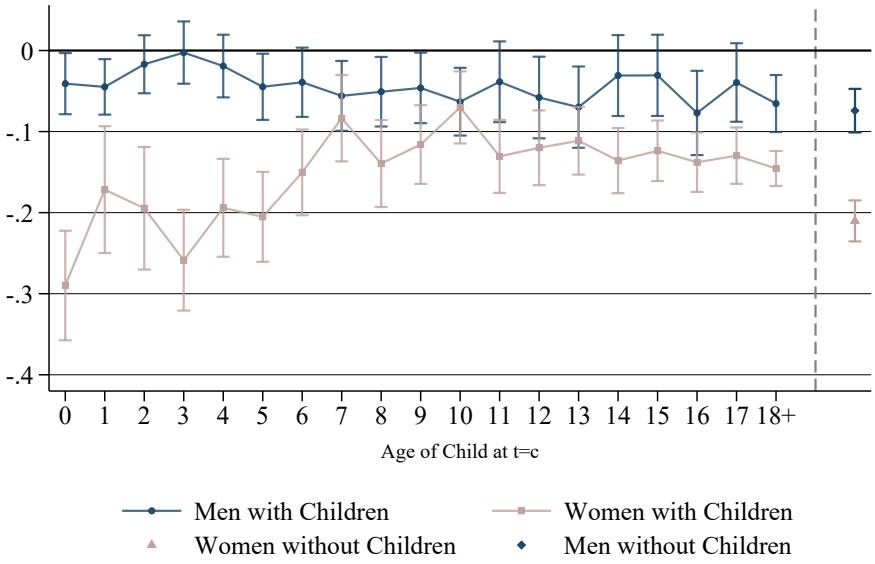
(b) Relative earnings

Notes: This figure presents heterogeneous effects of job displacement on labor market outcomes by young child status. Child < 8 indicates workers whose youngest child was younger than age 8 at the time of displacement, while No Child < 8 indicates workers with no child younger than age 8. The sample and fixed effects are identical to those in Figure 2. Each dot represents the estimated interaction coefficient between treatment status and event time, and capped lines indicate 95% confidence intervals. Standard errors are clustered at the company level.

Figure 4: Effects of Job Displacement on Labor Market Outcomes by Child Age



(a) Ever work



(b) Relative earnings

Notes: This figure presents estimated effects of job displacement on labor market outcomes by the age of the youngest child at the time of displacement, separately for women and men. The horizontal axis indicates the age of the youngest child in the event quarter, (t=c), with (18+) grouping workers whose youngest child is age 18 or older. Markers for workers without children are shown separately. Each marker represents the estimated treatment effect, and capped lines indicate 95% confidence intervals. Standard errors are clustered at the company level.

Tables

Table 1: Descriptive Statistics for Displaced and Non-Displaced Workers

	(1) Displaced Women	(2) Non-Displaced Women	SMD	(3) Displaced Men	(4) Non-Displaced Men	SMD
Number of Observations	24,979	24,979		21,885	21,885	
Panel A: Individual Characteristics						
Age ($c-1$)	42.102	42.091	-0.00	41.094	41.079	-0.00
	(5.792)	(5.768)		(5.676)	(5.680)	
Current tenure ($c-1$)	44.650	44.497	-0.00	54.669	54.549	-0.00
	(32.878)	(32.755)		(41.734)	(42.004)	
Total quarterly earnings ($c-4$)	527.866	533.294	0.02	941.585	942.974	0.00
	(231.089)	(242.054)		(431.754)	(430.766)	
Panel B: Child Status and Spouse Characteristics						
Have child before age 18	0.660	0.673	0.03	0.816	0.815	-0.00
	(0.474)	(0.469)		(0.387)	(0.388)	
Have child before age 8	0.217	0.219	0.00	0.418	0.419	0.00
	(0.412)	(0.413)		(0.493)	(0.493)	
Spouse quarterly earnings ($c-4$)	931.227	973.160	0.03	273.091	282.054	0.01
	(1,715.301)	(1,223.358)		(1,275.526)	(485.710)	
Spouse ever worked ($c-4$)	0.739	0.757	0.04	0.408	0.428	0.04
	(0.439)	(0.429)		(0.492)	(0.495)	
Panel C: Establishment Characteristics						
Revenue ($c-4$)	1,844.195	1,979.619	0.05	3,570.494	3,850.254	0.06
	(2,810.233)	(3,182.534)		(4,450.228)	(5,168.019)	
Employee 5–9	0.476	0.476	0	0.345	0.345	0
	(0.499)	(0.499)		(0.475)	(0.475)	
Employee 10–29	0.307	0.307	0	0.300	0.300	0
	(0.461)	(0.461)		(0.458)	(0.458)	
Employee 30–49	0.081	0.081	0	0.100	0.100	0
	(0.273)	(0.273)		(0.301)	(0.301)	
Employee 50–99	0.085	0.085	0	0.154	0.154	0
	(0.279)	(0.279)		(0.361)	(0.361)	
Employee 100–199	0.040	0.040	0	0.082	0.082	0
	(0.196)	(0.196)		(0.274)	(0.274)	
Employee 200–299	0.005	0.005	0	0.011	0.011	0
	(0.071)	(0.071)		(0.104)	(0.104)	
Employee 300–999	0.006	0.006	0	0.008	0.008	0
	(0.074)	(0.074)		(0.089)	(0.089)	

Notes: This table presents the pre-displacement characteristics of displaced and non-displaced workers, by gender. Age and tenure are reported as observed at quarter $t = c-1$, while the remaining firm-level variables are based on observations at $t = c-4$. Earnings and revenue variables have been adjusted for inflation and converted from nominal to real terms. Quarterly earnings are expressed in units of 10,000 KRW, and firm revenue in units of 1 million KRW. Standard deviations are in brackets.

Table 2: Descriptive Statistics of Displaced Workers in the Reference Quarter

	(1) Baseline Sample Women	(2) Reweighted Women	(3) Baseline Sample Men
Number of Observations	24,979	24,979	21,885
Panel A: Individual Characteristics			
Age $(c-1)$	42.102	41.177	41.094
	[5.792]	[6.029]	[5.676]
Current tenure $(c-1)$	44.650	52.929	54.669
	[32.878]	[40.926]	[41.734]
Total quarterly earnings $(c-4)$	527.866	792.031	941.585
	[231.089]	[310.383]	[431.754]
Panel B: Family and Spouse Characteristics			
Have child before age 18	0.660	0.629	0.816
	[0.474]	[0.483]	[0.387]
Have young child	0.217	0.248	0.418
	[0.412]	[0.432]	[0.493]
Spouse quarterly earnings $(c-4)$	931.227	1,092.578	273.091
	[1,715.301]	[4,280.752]	[1,275.526]
Spouse ever worked $(c-4)$	0.739	0.740	0.408
	[0.439]	[0.439]	[0.492]
Panel C: Establishment Characteristics			
Revenue $(c-4)$	1,844.195	2,959.796	3,570.494
	[2,810.233]	[3,555.093]	[4,450.228]
Employee 5–9	0.476	0.404	0.345
	[0.499]	[0.491]	[0.475]
Employee 10–29	0.307	0.309	0.300
	[0.461]	[0.462]	[0.458]
Employee 30–49	0.081	0.088	0.100
	[0.273]	[0.284]	[0.301]
Employee 50–99	0.085	0.128	0.154
	[0.279]	[0.334]	[0.361]
Employee 100–199	0.040	0.059	0.082
	[0.196]	[0.236]	[0.274]
Employee 200–299	0.005	0.005	0.011
	[0.071]	[0.068]	[0.104]
Employee 300–999	0.006	0.007	0.008
	[0.074]	[0.084]	[0.089]

Notes: This table presents the pre-displacement characteristics of displaced workers, by gender. Age, tenure, and quarterly earnings are observed at time $t = c-1$, while revenue and establishment-level variables are measured at $t = c-4$. Earnings and revenue are deflated to real terms. Column (2) contains displaced women reweighted to match men. Quarterly earnings are expressed in units of 10,000 KRW, and firm revenue in units of 1 million KRW. Standard deviations are in brackets.

Table 3: Effects of Job Displacement on Labor Market Outcomes

	(1) Ever Work	(2) Regular Employment	(3) Quarterly Earnings	(4) Relative Earnings
Panel A. Male				
Treated × 1(Year of displacement)	-0.096***	-0.176***	-159.7***	-0.163***
	(0.002)	(0.004)	(6.437)	(0.006)
Treated × 1(After 1year)	-0.054***	-0.157***	-110.7***	-0.100***
	(0.002)	(0.004)	(7.286)	(0.007)
Treated × 1(After 2year)	-0.041***	-0.135***	-87.6***	-0.072***
	(0.002)	(0.004)	(8.129)	(0.008)
Treated × 1(After 3year)	-0.027***	-0.114***	-69.1***	-0.053***
	(0.002)	(0.004)	(10.210)	(0.008)
Treated × 1(After 4year)	-0.027***	-0.107***	-66.1***	-0.054***
	(0.002)	(0.004)	(10.960)	(0.009)
Observations	1,050,480	1,050,480	1,050,480	1,050,480
Panel B. Female				
Treated × 1(Year of displacement)	-0.153***	-0.208***	-124.6***	-0.239***
	(0.002)	(0.003)	(2.647)	(0.005)
Treated × 1(After 1year)	-0.084***	-0.148***	-72.3***	-0.135***
	(0.003)	(0.003)	(3.227)	(0.006)
Treated × 1(After 2year)	-0.059***	-0.111***	-55.3***	-0.094***
	(0.003)	(0.004)	(3.574)	(0.007)
Treated × 1(After 3year)	-0.043***	-0.087***	-43.9***	-0.076***
	(0.003)	(0.004)	(3.813)	(0.007)
Treated × 1(After 4year)	-0.032***	-0.075***	-37.2***	-0.067***
	(0.003)	(0.004)	(3.983)	(0.008)
Observations	1,198,992	1,198,992	1,198,992	1,198,992
Panel C. Reweighted Female				
Treated × 1(Year of displacement)	-0.176***	-0.251***	-224.7***	-0.290***
	(0.005)	(0.006)	(9.608)	(0.009)
Treated × 1(After 1year)	-0.114***	-0.216***	-179.3***	-0.222***
	(0.006)	(0.008)	(12.100)	(0.011)
Treated × 1(After 2year)	-0.084***	-0.172***	-154.3***	-0.185***
	(0.006)	(0.008)	(12.640)	(0.011)
Treated × 1(After 3year)	-0.061***	-0.140***	-125.1***	-0.155***
	(0.006)	(0.009)	(13.380)	(0.012)
Treated × 1(After 4year)	-0.049***	-0.128***	-119.8***	-0.146***
	(0.006)	(0.009)	(13.360)	(0.013)
Observations	1,198,992	1,198,992	1,198,992	1,198,992
Constant	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes
Company Cluster	Yes	Yes	Yes	Yes

Notes: This table presents estimated effects of job displacement on labor market outcomes. "Year of displacement" refers to the four quarters including the quarter of displacement. "After year" refers to the four quarters following displacement (i.e., 1, 2, and 3 years after displacement). All models include calendar fixed effects and individual fixed effects. Standard errors clustered at the company level are reported in parentheses.

Table 4: Effects of Job Displacement on Labor Market Outcomes

	(1) Ever Work	(2) Regular Employment	(3) Quarterly Earnings	(4) Relative Earnings
Panel A. Baseline Female vs Male				
Post × Treat	-0.048***	-0.136***	-97.0***	-0.087***
	(0.002)	(0.004)	(7.451)	(0.007)
Post × Treat × Female	-0.024***	0.013***	32.0***	-0.033***
	(0.002)	(0.005)	(7.742)	(0.008)
Panel B. Reweighted Female vs Male				
Post × Treat	-0.048***	-0.136***	-97.0***	-0.087***
	(0.002)	(0.004)	(7.444)	(0.007)
Post × Treat × Female	-0.046***	-0.042***	-61.1***	-0.110***
	(0.005)	(0.007)	(12.620)	(0.011)
Constant	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes
Company Cluster	Yes	Yes	Yes	Yes
Observations	2,343,200	2,343,200	2,343,200	2,343,200

Notes: This table presents estimated effects of job displacement on labor market outcomes. Post is an indicator for all periods after job displacement. All models include calendar fixed effects and individual fixed effects. Standard errors clustered at the company level are reported in parentheses.

Table 5: Reweighted Effects of Job Displacement by the Presence of Young Children

	(1) Ever Work	(2) Regular Employment	(3) Quarterly Earnings	(4) Relative Earnings
Panel A. Have Young Child				
Post × Treat	-0.040***	-0.123***	-73.430***	-0.066***
	(0.002)	(0.005)	(10.140)	(0.009)
Post × Treat × Female	-0.100***	-0.137***	-125.000***	-0.195***
	(0.010)	(0.014)	(23.250)	(0.022)
Observations	729,775	729,775	729,775	729,775
Panel B. Have No Young Child				
Post × Treat	-0.054***	-0.146***	-114.000***	-0.102***
	(0.002)	(0.004)	(8.518)	(0.007)
Post × Treat × Female	-0.026***	-0.007	-31.590**	-0.074***
	(0.005)	(0.008)	(14.450)	(0.013)
Observations	1,613,425	1,613,425	1,613,425	1,613,425
Constant	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes
Company Cluster	Yes	Yes	Yes	Yes

Notes: This table presents reweighted estimated effects of job displacement on labor market outcomes separately for workers with and without young children. Young children are defined as children under age 8 at the time of displacement. Post is an indicator for all periods after job displacement. All models include calendar fixed effects and individual fixed effects. Standard errors clustered at the company level are reported in parentheses.

Appendix

A. Details on Matching Procedure

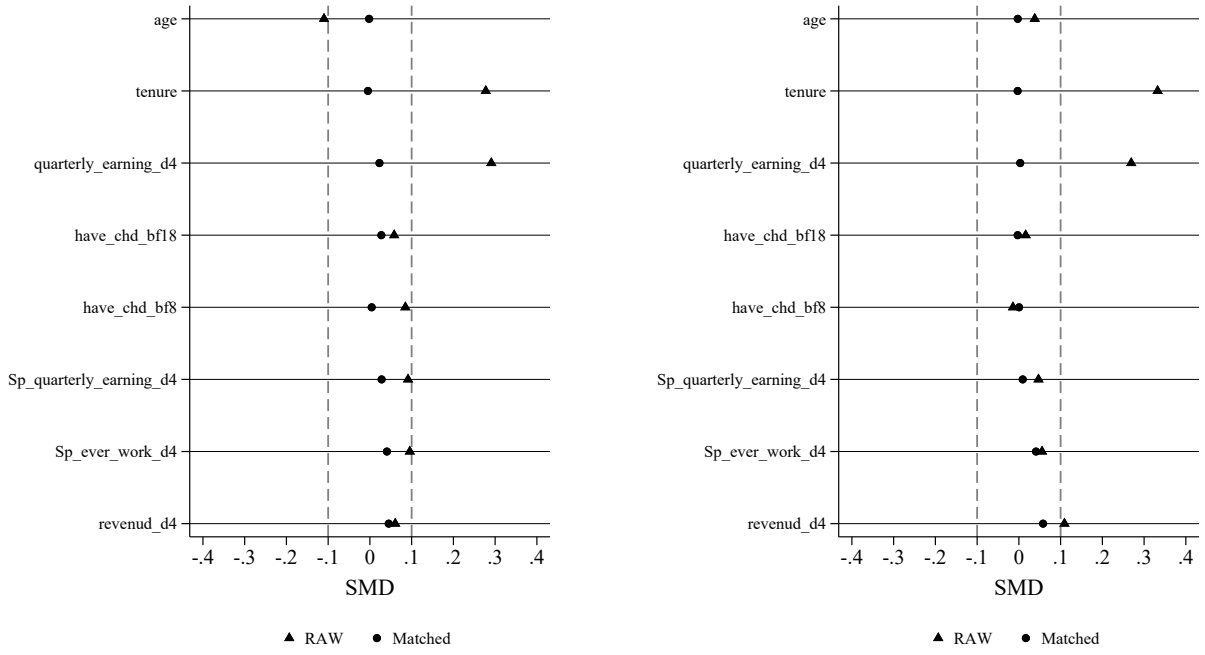
Matching is implemented using coarsened exact matching (CEM) without replacement. The procedure is conducted separately for men and women.

Age at $t = c - 1$ is grouped into five bins: 24–29, 30–34, 35–39, 40–44, and 45–50. Establishment size at $t = c - 4$ is classified into nine categories based on the Employer–Employee Data. Industry at $t = c - 4$ is coded into 21 categories following the Korean Standard Industrial Classification. Tenure at $t = c - 1$ is measured in months and grouped into two-year intervals.

In addition, observations in the top and bottom 1% of the revenue distribution and the top 1% of the quarterly earnings distribution are excluded separately for men and women to mitigate the influence of extreme values.

B. Additional Figures and Tables

Figure B.1: Standardized Mean Differences Before and After Matching



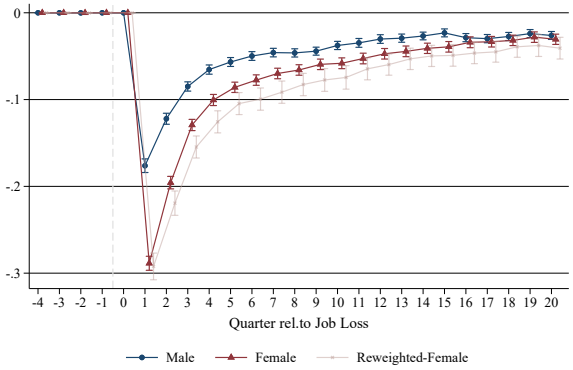
(a) Female Sample

(b) Male Sample

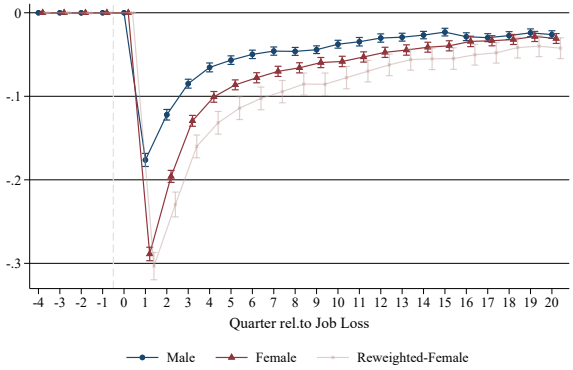
Notes: This figures present standardized mean differences (SMDs) for covariates before and after matching. Black triangles indicate the raw sample, and black circles indicate the matched sample. The dashed vertical lines correspond to ± 0.1, a commonly used threshold for acceptable balance.

Figure B.2: Key Drivers of Reweighted Displacement Losses in Ever Work

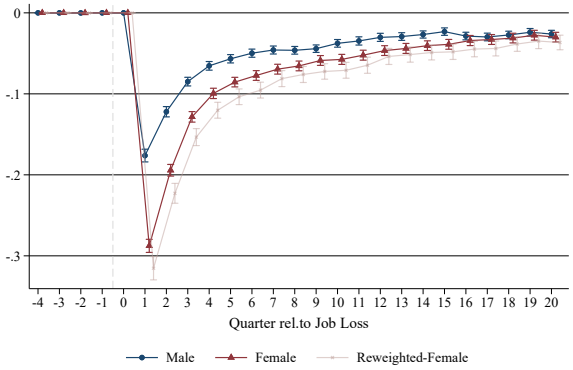
(a) Excluding Age



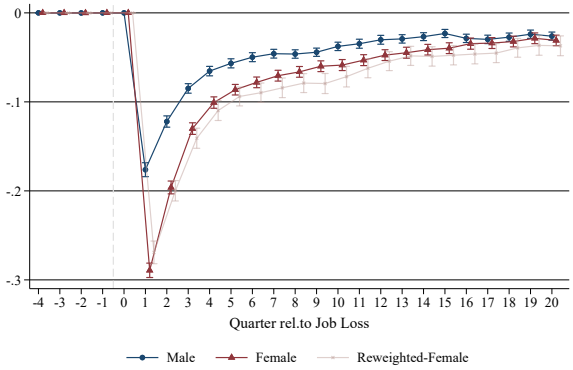
(b) Excluding Firm Size



(c) Excluding Pre-Displacement Earnings



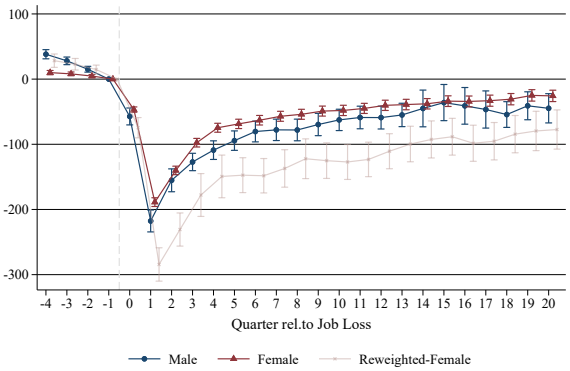
(d) Excluding Industry



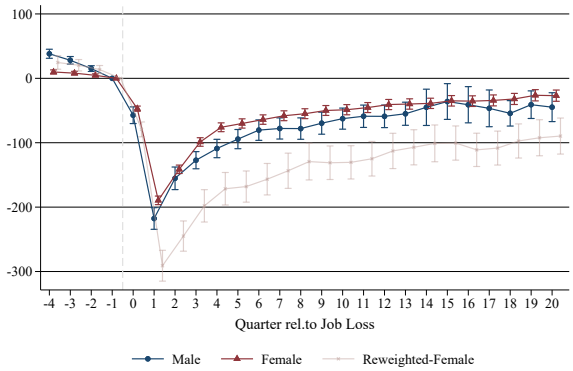
Notes: This figures present reweighted dynamic difference-in-differences estimates of the effects of job displacement on ever work. Female workers are reweighted to match the pre-displacement characteristics of male workers, excluding one covariate from the reweighting procedure in each panel. Panel (a) excludes age, Panel (b) excludes firm size, Panel (c) excludes pre-displacement earnings and Panel (c) excludes industry. The remaining reweighting covariates are the same as in the baseline reweighting specification. The specification includes calendar-time fixed effects and individual fixed effects. Each dot represents the estimated interaction coefficient between treatment status and event time, and capped lines indicate 95% confidence intervals. Standard errors are clustered at the company level.

Figure B.3: Key Drivers of Reweighted Displacement Losses in Quarterly Earnings

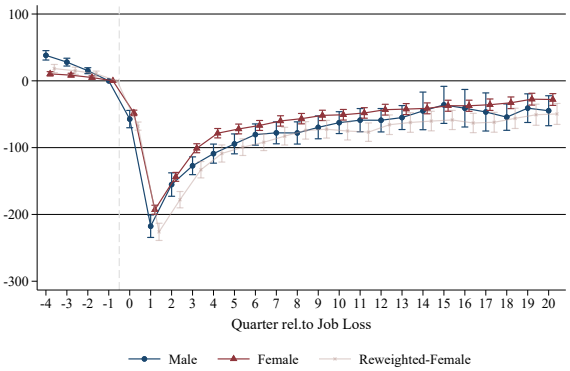
(a) Excluding Age



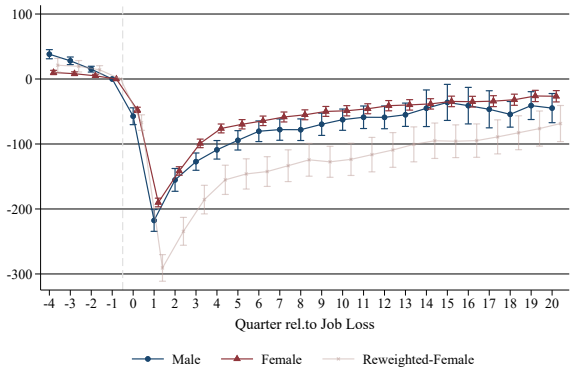
(b) Excluding Firm Size



(c) Excluding Pre-Displacement Earnings

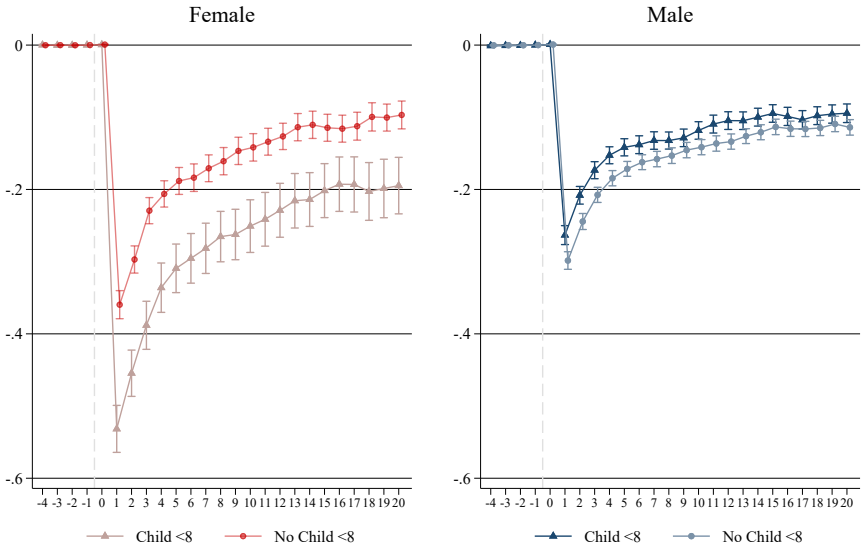


(d) Excluding Industry

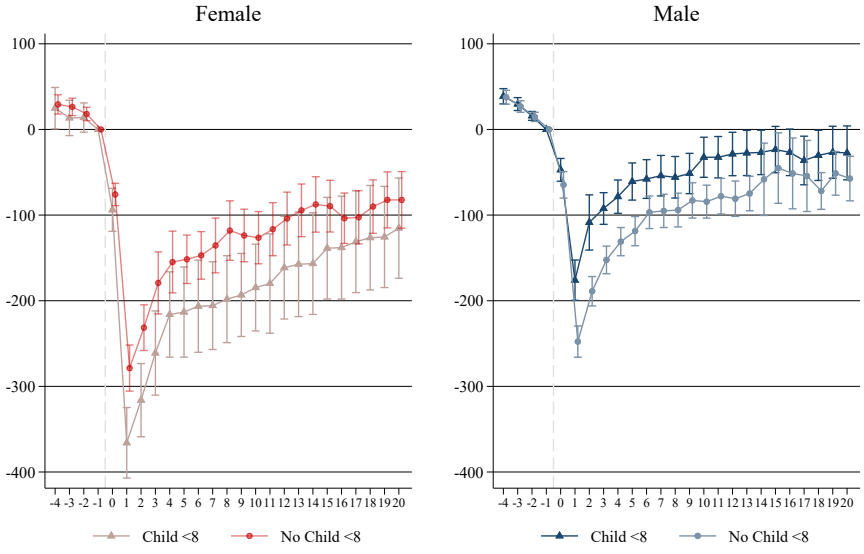


Notes: This figures present reweighted dynamic difference-in-differences estimates of the effects of job displacement on quarterly earnings. Female workers are reweighted to match the pre-displacement characteristics of male workers, excluding one covariate from the reweighting procedure in each panel. Panel (a) excludes age, Panel (b) excludes firm size, Panel (c) excludes pre-displacement earnings, and Panel (d) excludes industry. The remaining reweighting covariates are the same as in the baseline reweighting specification. The specification includes calendar-time fixed effects and individual fixed effects. Each dot represents the estimated interaction coefficient between treatment status and event time, and capped lines indicate 95% confidence intervals. Standard errors are clustered at the company level.

Figure B.4: Effects of Job Displacement on Labor Market Outcomes by the Presence of Young Children



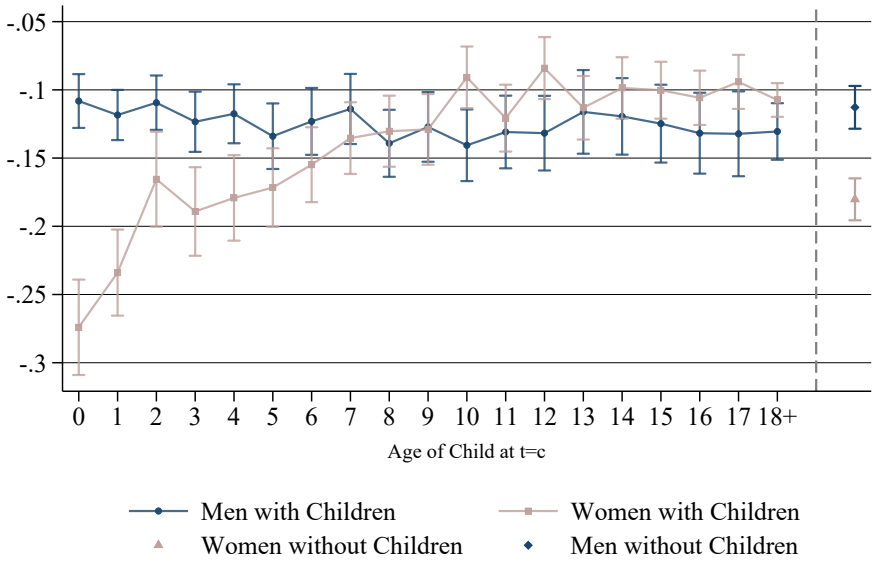
(a) Regular Employment



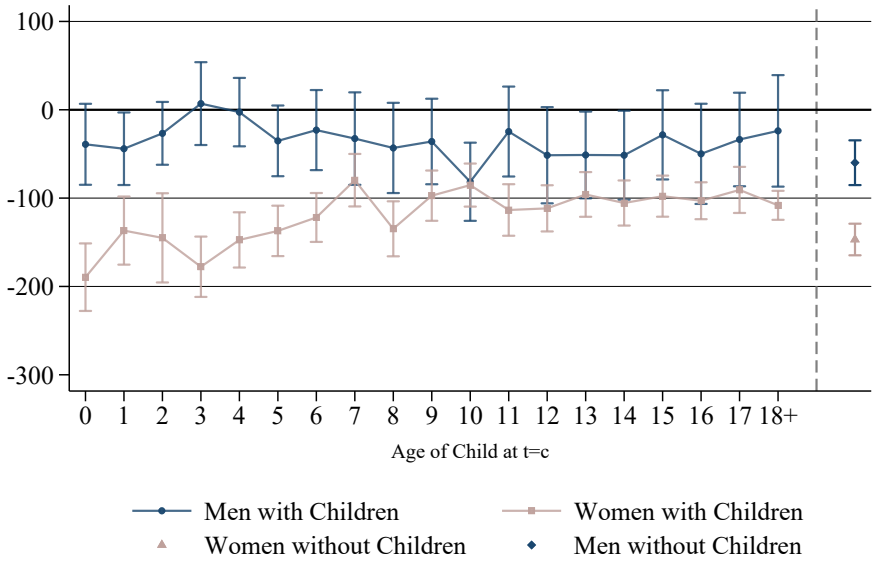
(b) Quarterly Earnings

Notes: This figures present heterogeneous effects of job displacement on labor market outcomes by young child status. Child < 8 indicates workers whose youngest child was younger than age 8 at the time of displacement, while No Child < 8 indicates workers with no child younger than age 8. The sample and fixed effects are identical to those in Figure 2. Each dot represents the estimated interaction coefficient between treatment status and event time, and capped lines indicate 95% confidence intervals. Standard errors are clustered at the company level.

Figure B.5: Effects of Job Displacement on Labor Market Outcomes by Child Age



(a) Regular Employment



(b) Quarterly Earnings

Notes: This figures present estimated effects of job displacement on labor market outcomes by the age of the youngest child at the time of displacement, separately for women and men. The horizontal axis indicates the age of the youngest child in the event quarter, (t=c), with (18+) grouping workers whose youngest child is age 18 or older. Markers for workers without children are shown separately. Each marker represents the estimated treatment effect, and capped lines indicate 95% confidence intervals. Standard errors are clustered at the company level.

Table B.1: Descriptive Characteristics of Single- and Multi-Establishment Firms

	Multi-company Firms	Single-company Firms
A. Coverage in 2016		
Share of firms	0.02	0.98
Share of workers	0.43	0.57
B. Characteristics		
Average number of workers	127	6
Age	43	49
Annual income	2,373	1,693
Tenure	40	67
Regular worker share	0.57	0.18
Daily worker share	0.16	0.08
Full-time worker share	0.41	0.10
Married share	0.77	0.82
Married share(83-95)	0.21	0.24
Revenue	22,176	401
C.Industry Shares		
Manufacturing (C)	0.22	0.09
Wholesale and retail trade (G)	0.10	0.25
Transportation and storage (H)	0.03	0.11
Accommodation and food service (I)	0.05	0.15
Real estate activities (L)	0.05	0.11

Notes: This table compares descriptive characteristics of single-establishment and multi-establishment firms in 2016. Single-establishment firms refer to firms with one establishment, while multi-establishment firms refer to firms with more than one establishment. In Panel A, the firm share is calculated using firms as the unit of observation, while the worker share is calculated using workers as the unit of observation. Panels B and C use the firm as the unit of observation. Worker characteristics in Panel B are measured as firm-level averages or shares among workers within each firm. Annual income is expressed in units of 10,000 KRW, and revenue is expressed in units of 1 million KRW.

Table B.2: Descriptive Statistics for Displaced and Non-Displaced Workers

	(1) Displaced Women	(2) Non-Displaced Women	SMD	(3) Displaced Men	(4) Non-Displaced Men	SMD
A	0.000	0.000	0	0.001	0.001	0
	(0.018)	(0.018)		(0.035)	(0.035)	
B	0.000	0.000	0	0.000	0.000	0
	(0.000)	(0.000)		(0.000)	(0.000)	
C	0.182	0.182	0	0.448	0.448	0
	(0.386)	(0.386)		(0.497)	(0.497)	
D	0.000	0.000	0	0.000	0.000	0
	(0.009)	(0.009)		(0.010)	(0.010)	
E	0.000	0.000	0	0.003	0.003	0
	(0.021)	(0.021)		(0.055)	(0.055)	
F	0.026	0.026	0	0.081	0.081	0
	(0.159)	(0.159)		(0.273)	(0.273)	
G	0.074	0.074	0	0.102	0.102	0
	(0.262)	(0.262)		(0.303)	(0.303)	
H	0.020	0.020	0	0.031	0.031	0
	(0.141)	(0.141)		(0.173)	(0.173)	
I	0.000	0.000	0	0.000	0.000	0
	(0.000)	(0.000)		(0.000)	(0.000)	
J	0.007	0.007	0	0.029	0.029	0
	(0.083)	(0.083)		(0.168)	(0.168)	
K	0.000	0.000	0	0.001	0.001	0
	(0.019)	(0.019)		(0.036)	(0.036)	
L	0.005	0.005	0	0.007	0.007	0
	(0.074)	(0.074)		(0.083)	(0.083)	
M	0.027	0.027	0	0.044	0.044	0
	(0.162)	(0.162)		(0.204)	(0.204)	
N	0.075	0.075	0	0.088	0.088	0
	(0.263)	(0.263)		(0.284)	(0.284)	
O	0.000	0.000	0	0.000	0.000	0
	(0.000)	(0.000)		(0.007)	(0.007)	
P	0.016	0.016	0	0.008	0.008	0
	(0.125)	(0.125)		(0.090)	(0.090)	
Q	0.477	0.477	0	0.047	0.047	0
	(0.499)	(0.499)		(0.211)	(0.211)	
R	0.002	0.002	0	0.005	0.005	0
	(0.046)	(0.046)		(0.068)	(0.068)	
S	0.023	0.023	0	0.047	0.047	0
	(0.150)	(0.150)		(0.211)	(0.211)	
Number of Observations	24,979	24,979		21,885	21,885	

Notes: This table presents the pre-displacement characteristics of displaced and non-displaced workers, by gender. In addition to Table 1, this table includes industry classification variables in the descriptive statistics. Standard deviations are in brackets.

Table B.3: Descriptive Statistics of Displaced Workers in the Reference Quarter

	(1) Baseline Sample Women	(2) Reweighted Women	(3) Baseline Sample Men
Number of Observations	24,979	24,979	21,885
A	0.000	0.001	0.001
	[0.018]	[0.029]	[0.035]
B	0.000	0.000	0.000
	[0.000]	[0.000]	[0.000]
C	0.182	0.375	0.448
	[0.386]	[0.484]	[0.497]
D	0.000	0.000	0.000
	[0.009]	[0.006]	[0.010]
E	0.000	0.001	0.003
	[0.021]	[0.035]	[0.055]
F	0.026	0.077	0.081
	[0.159]	[0.266]	[0.273]
G	0.074	0.133	0.102
	[0.262]	[0.339]	[0.303]
H	0.020	0.030	0.031
	[0.141]	[0.170]	[0.173]
I	0.000	0.000	0.000
	[0.000]	[0.000]	[0.000]
J	0.007	0.023	0.029
	[0.083]	[0.149]	[0.168]
K	0.000	0.000	0.001
	[0.019]	[0.021]	[0.036]
L	0.005	0.008	0.007
	[0.074]	[0.087]	[0.083]
M	0.027	0.060	0.044
	[0.162]	[0.237]	[0.204]
N	0.075	0.093	0.088
	[0.263]	[0.291]	[0.284]
O	0.000	0.000	0.000
	[0.000]	[0.000]	[0.007]
P	0.016	0.013	0.008
	[0.125]	[0.112]	[0.090]
Q	0.477	0.060	0.047
	[0.499]	[0.237]	[0.211]
R	0.002	0.006	0.005
	[0.046]	[0.074]	[0.068]
S	0.023	0.048	0.047
	[0.150]	[0.213]	[0.211]

Notes: This table presents the pre-displacement characteristics of displaced workers, by gender. In addition to Table 2, this table includes industry classification variables in the descriptive statistics. Column (2) contains displaced women reweighted to match men. Standard deviations are in brackets.

Table B.4: Effects of Job Displacement on Labor Market Outcomes

	(1) Ever Work	(2) Regular Employment	(3) Quarterly Earnings	(4) Relative Earnings
Panel A. Male				
Treated × 1(Post)	-0.048***	-0.136***	-97.0***	-0.087***
	(0.002)	(0.004)	(7.422)	(0.007)
Observations	1,094,250	1,094,250	1,094,250	1,094,250
Panel B. Female				
Treated × 1(Post)	-0.072***	-0.123***	-65.0***	-0.119***
	(0.002)	(0.003)	(2.857)	(0.005)
Observations	1,248,950	1,248,950	1,248,950	1,248,950
Panel C. Reweighted Female				
Treated × 1(Post)	-0.095***	-0.179***	-158.1***	-0.196***
	(0.005)	(0.007)	(10.650)	(0.010)
Observations	1,248,950	1,248,950	1,248,950	1,248,950
Constant	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes
Company Cluster	Yes	Yes	Yes	Yes

Notes: This table presents estimated effects of job displacement on labor market outcomes. Post is an indicator for all periods after job displacement. All models include calendar fixed effects and individual fixed effects. Standard errors clustered at the company level are reported in parentheses.